Alpha Signal Construction and Backtesting Report

1. Introduction

This project analyzes historical stock price data of Apple Inc. (AAPL) to construct and evaluate alpha signals based on momentum and volatility.

The objective is to determine whether these signals can generate excess returns compared to a buy-and-hold strategy.

2. Data Cleaning & Preparation

The dataset initially contained inconsistent headers and incorrect data types.

After cleaning, all variables were converted into appropriate formats and sorted chronologically.

No missing values or duplicate timestamps were found, ensuring reliability for time-series analysis.

3. Price Behavior Analysis

The price chart shows a strong long-term upward trend.

Several sharp corrections are visible, corresponding to market shocks.

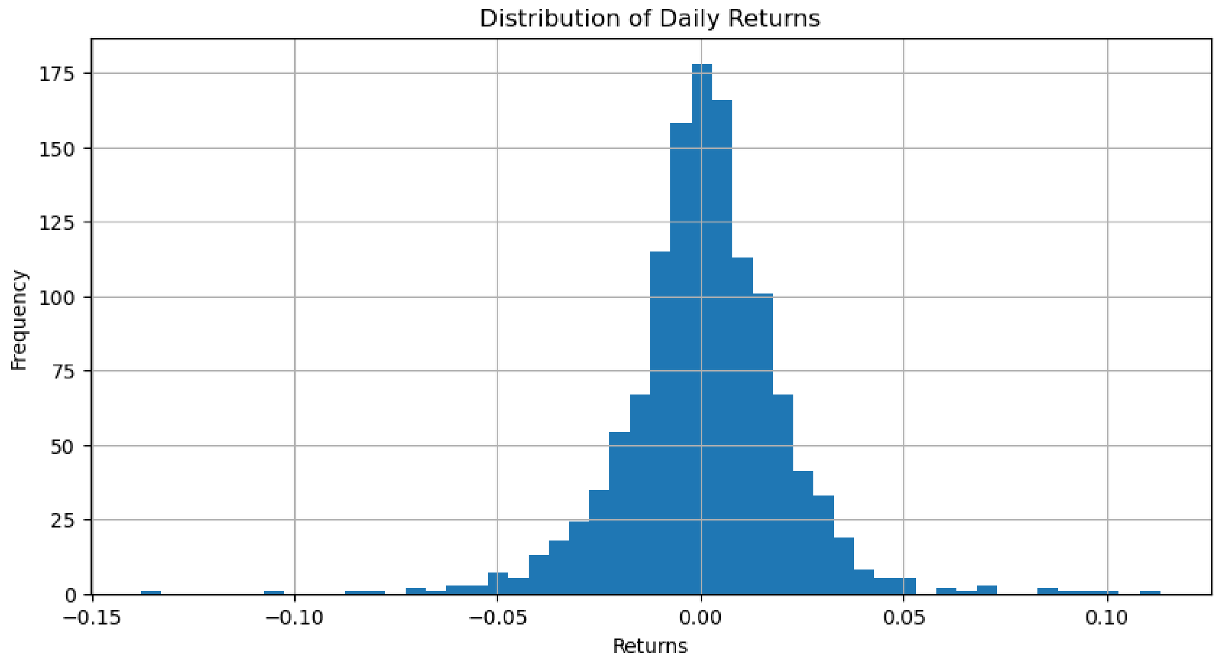
This indicates growth with intermittent volatility, typical of large-cap stocks.

4. Returns Distribution

Returns are centered around zero, which is expected in financial markets.

The distribution shows fat tails, indicating a higher probability of extreme movements.

This suggests non-normality in returns.

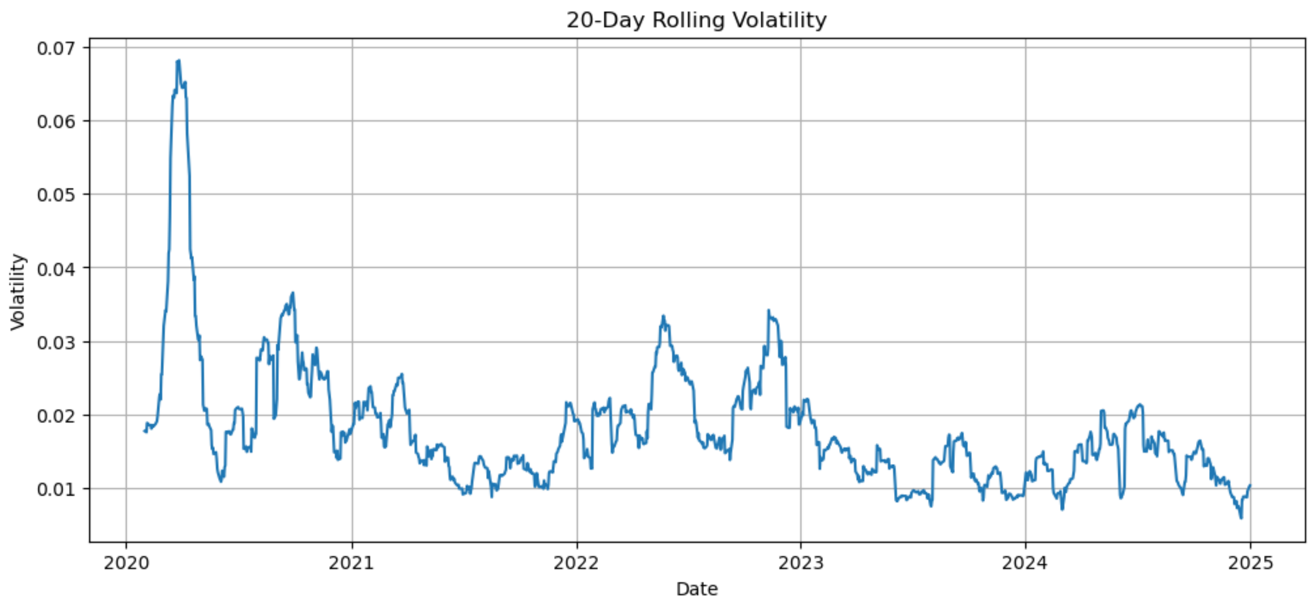


5. Volatility Behavior

The rolling volatility plot shows clustering behavior.

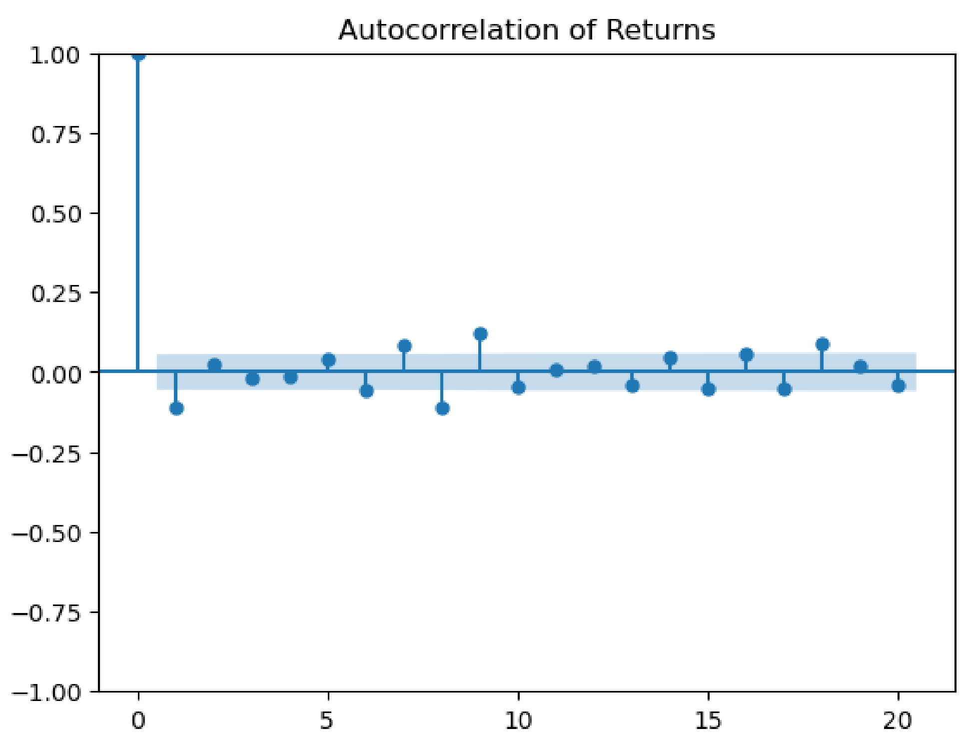
Periods of high volatility are followed by high volatility.

This confirms the presence of volatility persistence.

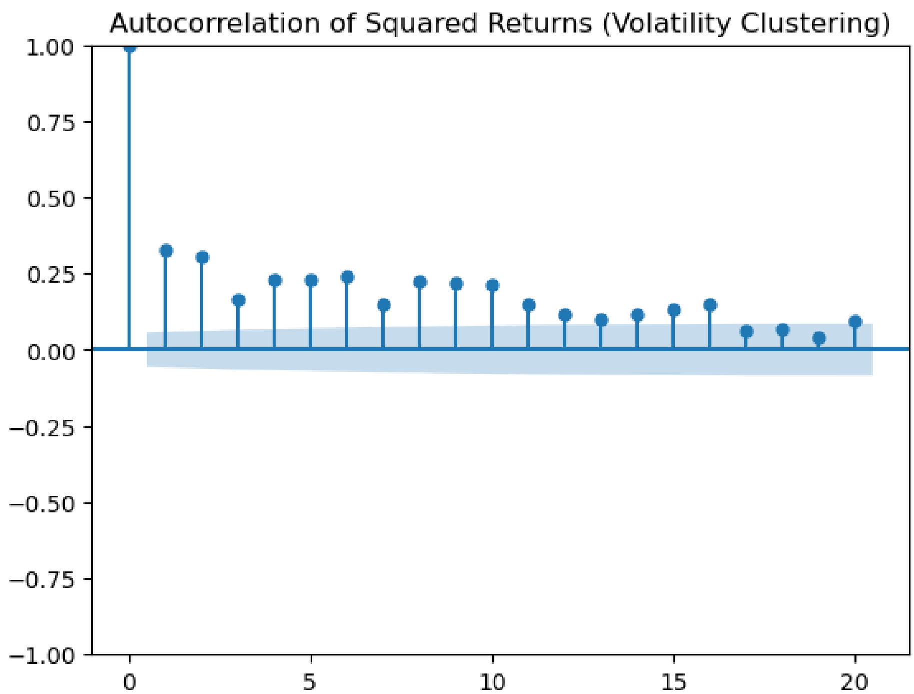


6. Statistical Analysis

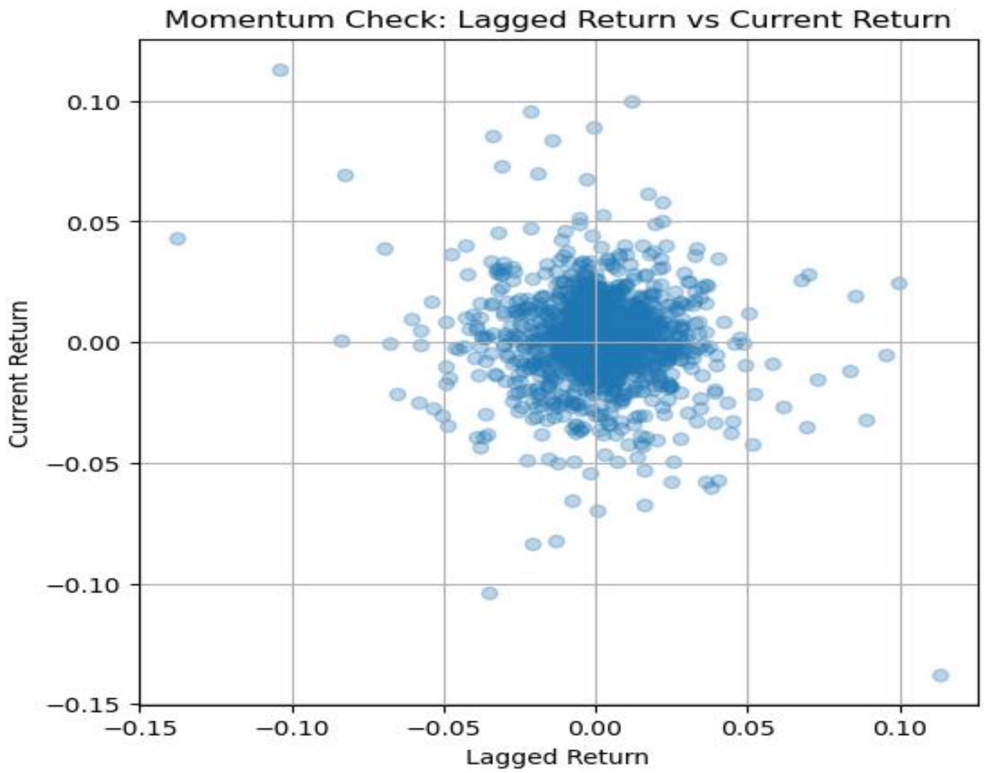
Autocorrelation of returns is weak and slightly negative (~ -0.11), indicating near market efficiency.



Squared returns show strong autocorrelation (~ 0.33), confirming volatility clustering.



Momentum persistence is weak and slightly negative, suggesting short-term reversal rather than trend continuation.



7. Feature Engineering

Daily log returns, rolling volatility, and momentum indicators were computed.

These features help capture return dynamics, risk levels, and trend behavior.

The transformed dataset is suitable for building trading strategies.

8. Alpha Signal Construction

Momentum signal: Buy when momentum is positive.

Volatility signal: Buy during low volatility periods.

Combined signal: Buy only when both conditions are satisfied.

This approach aims to capture trends while avoiding risky periods.

9. Backtesting

Signals were shifted to avoid look-ahead bias.

Strategy returns were calculated using lagged signals.

Performance was compared against a buy-and-hold benchmark.

10. Performance Evaluation

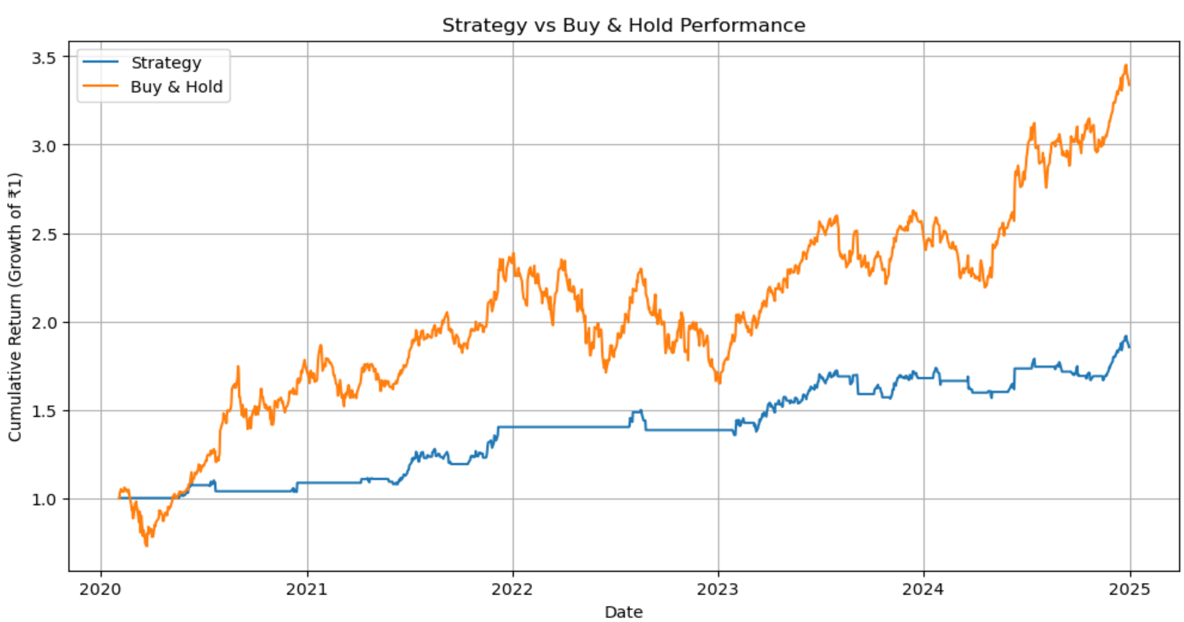
Strategy Mean Return: 0.00050 vs Buy-and-Hold: 0.00097

Strategy Volatility: 0.00776 vs Buy-and-Hold: 0.01997

Strategy Sharpe Ratio: 0.064 vs Buy-and-Hold: 0.049

Strategy Cumulative Return: 85% vs Buy-and-Hold: 233%

The strategy underperforms in total returns but performs better on a risk-adjusted basis.



11. Key Insights

Markets exhibit weak return predictability.

Volatility clustering is strong and useful for risk filtering.

Momentum is weak and unreliable in this dataset.

Volatility filtering improves stability but reduces returns.

12. Conclusion

The strategy demonstrates the trade-off between risk and return.

While it reduces volatility and improves Sharpe ratio, it misses high-return opportunities.

A balanced approach is required for optimal performance.